T2FGAN: Improving Semantic Embedding in Text-to-Face Generation for High-Resolution Images using A Multi-Stage Hybrid GAN

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Abstract: One of the most significant and complex challenges in deep learning is generating visuals from textual descriptions, with text-to-face generation being a particularly specialized area that focuses on creating facial images based on text input. Current text-to-face synthesis models struggle to balance semantic accuracy with photorealistic detail, due to limitations in handling the complex semantic space of facial attributes such as identity, expression, age, and ethnicity, often resulting in low-resolution or visually bland images that fail to faithfully reflect rich textual descriptions. To address these fundamental limitations, we propose T2FGAN, a novel three-layered GAN architecture that integrates DCGAN, SRGAN, and attention mechanisms to achieve high-quality text-to-face generation while maintaining both semantic fidelity and visual realism. Our model consists of three generators and two discriminators: the first generator, based on DCGAN, performs initial text-to-face translation using Sentence-BERT embeddings; the second generator incorporates an attention mechanism to refine facial details while preserving semantic alignment; and the third generator, built on SRGAN, enhances resolution to 256x256 pixels for photorealistic output. This multi-stage layered approach effectively combines semantic preservation, detail enhancement, and resolution improvement to deliver realistic and personalized face images from textual descriptions, representing a significant advancement in addressing the core challenge of balancing semantic accuracy with photorealistic detail in text-to-face synthesis.

Keywords: Text-to-Face, GAN, Deep Learning, Multi-Stage Generation, Attention Mechanism, High-Resolution Synthesis, T2FGAN, DCGAN, SRGAN, Sentence-BERT

1 Introduction

Text-to-image synthesis has become a prominent research area in deep learning, where the goal is to generate realistic images based on textual descriptions. One of the more specialized tasks in this field is *text-to-face generation*, where the challenge is to create highly detailed human faces from descriptive text input. While significant progress has been made in generating images of objects like animals, birds, and flowers, generating human faces is inherently more complex due to the fine-grained details and the intricate relationships between facial features. Additionally, faces are more context-sensitive than other objects, requiring precise control over features such as age, gender, ethnicity, and expression, making text-to-face generation a particularly challenging problem [1].

Current text-to-face synthesis models struggle to balance semantic accuracy with photorealistic detail due to limitations in handling the complex semantic space of facial attributes (e.g., identity, expression, age, ethnicity). Natural language is inherently abstract, whereas images are concrete. This discrepancy poses a significant challenge, especially when utilizing multiple descriptions to generate accurate images. Existing models often generate low-resolution or visually bland images, failing to faithfully reflect rich textual descriptions. As a sub-domain of text-to-image synthesis, text-to-face generation has huge potential in the public safety domain. Due to the lack of datasets, there is a scarcity of research focusing specifically on text-to-face synthesis [2], highlighting the need for more sophisticated approaches that can handle the nuanced requirements of facial generation.

Recent advancements in Generative Adversarial Networks (GANs) have significantly improved text-to-image synthesis [3]. Notable models such as StackGAN [4] utilize a two-stage generation process to create high-quality images by progressively refining low-resolution outputs. DCGAN employs deep convolutional networks to enhance image quality, focusing on both realism and fine-grained detail. Additionally, AttnGAN [5] incorporates attention mechanisms that dynamically focus on key textual elements, enhancing the model's ability to generate images that are semantically aligned with input descriptions. SRGAN [6] further improves image quality by upscaling low-resolution images, generating high-resolution outputs with enhanced detail, making it an essential tool for applications requiring fine-grained visual fidelity. More recently, StyleGAN [7] and BigGAN have significantly advanced the fidelity and resolution of generated images.

To better map textual inputs to visual features, various text embedding models have been developed. Early models like Word2Vec and GloVe [8] represent words as vectors in a high-dimensional space, capturing simple relationships between words. However, these models are static and fail to account for context. More advanced models, such as BERT [9] and Sentence-BERT [10], use transformer-based architectures to provide contextualized word embeddings, which improve semantic understanding by capturing interactions between words based on sentence context. These text embedding models, alongside generative techniques, have enabled the synthesis of increasingly realistic and contextually accurate images from textual descriptions.

While GANs have seen great success in generating images from text, generating human faces is significantly more challenging due to the need for high fidelity in facial details. Faces require precise control over features like age, gender, ethnicity, and expression, all of which are intricately linked. Even minor changes in facial attributes can result in a loss of realism, making text-to-face generation a more complex and demanding problem [11].

To address these limitations, we propose **T2FGAN**, a novel approach to text-to-face (T2F) generation using a three-layered GAN architecture with attention mechanisms and high-resolution image generation. Our solution tackles the core problem of balancing semantic accuracy with photorealistic detail through a multi-stage approach. The model consists of three generators: The first, based on DCGAN [1], creates an initial low-resolution face from text embeddings. The second refines the face using an attention mechanism [5] to focus on key facial features, ensuring semantic alignment with the input description. The third, based on SRGAN [6], enhances the resolution to 256×256 pixels for photorealistic output. Two discriminators evaluate both the semantic alignment with the text and the visual quality of the generated face, ensuring realistic and contextually accurate results. Text embeddings are obtained using Sentence-BERT [10] to capture full context, enabling precise face generation that maintains both semantic fidelity and visual realism.

The paper is organized as follows. Section 2 provides a literature survey, reviewing existing methods in text-to-image and text-to-face generation, as well as the various text embedding models used to map textual inputs to visual features. Section 3 introduces our proposed model, T2FGAN, detailing the architecture and explaining the roles of DCGAN, SRGAN, the attention mechanism, and embedding models such as Sentence-BERT. In Section 4, we present the results of our experiments, comparing T2FGAN's performance with existing models. Section 5 discusses the future scope of text-to-face generation, highlighting potential improvements and avenues for further research. Finally, Section 6 concludes the paper, summarizing our contributions and key findings.

2 Literature Survey

The concept of Generative Adversarial Networks (GANs) was introduced by Goodfellow et al. [12] in 2014, revolutionizing image synthesis through a dual-network system: a generator that produces synthetic images and a discriminator that evaluates their authenticity. This adversarial training framework has facilitated significant advances in generating high-quality images across various domains. For instance, GANs have been successfully applied in the medical domain for synthesizing retinal fundus images to address data scarcity challenges, as demonstrated in ReFIGG [13].

Extending GANs, the task of text-to-image (T2I) generation focuses on synthesizing images from textual descriptions. This process demands not only semantic understanding but also precise visual synthesis capabilities. Within T2I, text-to-face (T2F) generation emerges as a specialized problem, wherein models must generate facial images that accurately reflect detailed textual inputs. Given the fine-grained control required over facial attributes, generating realistic human faces remains particularly challenging. Early contributions such as Text2FaceGAN [14] explored GAN-based solutions for face generation from descriptive text.

To bridge the gap between textual semantics and visual features, embedding models have been extensively employed. These models capture the latent meaning of text, facilitating better attribute mapping during image generation. Early approaches relied on static word embeddings, such as Word2Vec and GloVe, which represent words in a fixed high-dimensional space, thereby capturing simple relationships. However, static embeddings lack contextual awareness, limiting their effectiveness for complex tasks. In contrast, modern methods like BERT [9] and Sentence-BERT [10] employ transformer architectures to generate contextualized embeddings that adapt dynamically based on sentence structure, substantially improving semantic representation.

Table 1 presents a comparative analysis of key embedding models, highlighting their architectures, contextual understanding, embedding sizes, training objectives, computational speed, and memory consumption. This comparison underscores the trade-offs between simplicity, performance, and computational cost when selecting embedding models for text-to-image generation tasks.

Recent surveys [3] categorize T2I approaches into four main paradigms: semantic enhancement, progressive networks, attention mechanisms, and additional signal incorporation. Semantic enhancement techniques focus on improving the interpretation of textual inputs to achieve better image quality. Progressive networks generate images in multiple stages, gradually refining

Aspect	Word2Vec	GloVe [8]	BERT	Sentence-BERT
Architecture	Shallow neural network (Skip-gram or CBOW)	Matrix factorization of word pairs	Transformer-based (Bidirectional)	Siamese network using BERT
Contextual Understanding	Static (same for each word)	Static (same for each word)	Contextual (changes based on sentence)	Contextual for sentences, fine-tuned for similarity
Embedding Size	50–300 dimensions	50, 100, 200, 300 dimensions	768 (Base), 1024 (Large) per token	768 (Base), 1024 (Large) per sentence
Training Objective	Predict context or target word	Minimize error in word co-occurrence	Masked Language Model (MLM) and Next Sentence Prediction (NSP)	Fine-tuned for sentence similarity
Speed	Fast	Fast	Slow (due to large model)	Slow (due to large model)
Memory Consumption	Low (small embeddings)	Low (small embeddings)	High (due to large model)	High (due to large model)

Table 1 Comparative analysis of key text embedding models—Word2Vec, GloVe, BERT, and Sentence-BERT—highlighting differences in architecture, contextual understanding, embedding dimensions, training objectives, speed, and memory requirements.

them from low to high resolution. Attention-based models dynamically focus on salient parts of the text to capture intricate details, while additional signal-based methods incorporate spatial or semantic cues to enhance the generated output.

Several notable architectures exemplify these paradigms. StackGAN [4] adopts a two-stage generation process, producing low-resolution images initially and refining them in a second stage to enhance detail. DCGAN employs deep convolutional layers to generate high-fidelity images, while AttnGAN [5] integrates attention mechanisms to ensure better semantic alignment between text and generated images.

Furthermore, GAN variants designed for image enhancement have gained traction. SRGAN [15] achieves super-resolution image generation with photorealistic textures. CycleGAN [16] enables unpaired image-to-image translation, while Pix2Pix [17] demonstrates effectiveness in conditional image generation tasks. The introduction of Wasserstein GAN (WGAN) [18] further improved training stability by optimizing the Earth Mover's distance instead of the Jensen–Shannon divergence.

Recent advancements in text-to-face (T2F) generation, such as FTGAN [19], have emphasized improving facial detail synthesis through refined adversarial training strategies. Despite these developments, significant challenges remain—particularly in generating facial images that are not only photorealistic but also semantically aligned with complex and nuanced textual descriptions. Additionally, a study on enhanced image super-resolution using a hierarchical generative adversarial network [20] presents a novel approach to improving image resolution through hierarchical GAN architectures, which is particularly beneficial for applications requiring high-fidelity outputs. This underscores the ongoing efforts to refine image synthesis techniques, which are critical for achieving realism in T2F generation. Future work should explore the integration of advanced word embeddings, innovative GAN architectures, and multimodal signals to enhance the realism, diversity, and semantic accuracy of the generated faces.

3 T2FGAN

T2FGAN comprises three generation stages and two evaluation discriminators:

1. **Generetor 1 (DCGAN)**: Synthesizes an initial 64×64 face from noise and Sentence-BERT embeddings.
2. **Generator 2 (Attention-GAN)**: Refines facial details guided by attention over textual embeddings.
3. **Generator 3 (SRGAN)**: Upscales the refined face to 256×256 with added texture fidelity.
4. **Discriminator 1**: Evaluates semantic relevance between generated image and text.

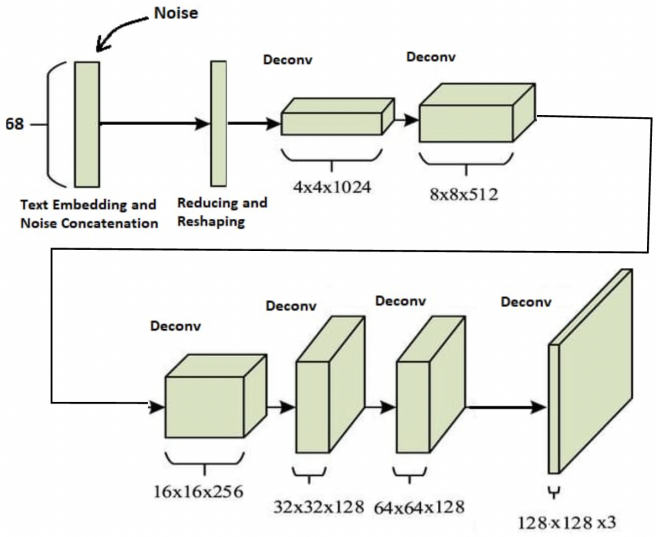


Figure 1: Generator 1: A 768-dimensional text embedding is combined with 100-dimensional noise and upsampled using transposed convolutions, increasing resolution from (4×4) to (128×128) to generate an RGB image.

5. **Discriminator 2**: Judges photorealism and texture quality.

Losses include adversarial loss, perceptual loss, and attention-weighted semantic consistency loss. Training is conducted in three phases using Adam optimizer.

3.1 Generator 1

The Generator 1 is based on the generator of DCGAN . It begins by projecting a 768-dimensional text embedding to a reduced dimension using a linear layer with batch normalization and activation, preparing it for combination with noise. This projected embedding is concatenated with a noise vector of dimension 100, creating a tensor with an expanded channel dimension.
The concatenated tensor is reshaped and then undergoes several upsampling steps using transposed convolutions as shown in Figure 1. Each step doubles the spatial dimensions and reduces the number of channels, gradually refining the resolution from (4x4) to (128x128). The first transposed convolution expands to 1024 channels at (4x4), and successive layers continue upsampling to (8x8), (16x16), (32x32), (64x64), and finally to (128x128). At each step, batch normalization and activation functions ensure stability and non-linearity.
The final output has 3 channels (for RGB images), yielding an image of size (128x128).

3.2 Discriminator 1

The discriminator 1 network is designed to differentiate real image-text pairs from generated ones. It consists of two main encoding paths: an image encoder and a text encoder, which is feed into a combined feature map before classification.

The image encoder takes an image input of shape (3 x 128 x 128) and applies a series of five convolutional layers, each progressively reducing spatial dimensions by a factor of two while increasing feature depth. Specifically, the first two layers maintain the feature depth of 128 but scale spatial dimensions down to (32x32). The next three layers increase feature depth sequentially to 256, 512, and 1024, while reducing spatial dimensions to (4x4), resulting in a feature map of (1024x4x4). In parallel, the text encoder processes a text embedding of dimension 768, reducing it to a 256 feature representation via a fully connected linear layer. To align this reduced text feature with the image feature map's spatial dimensions, the text encoding is replicated across a (4x4) grid, yielding dimensions (256x4x4). These replicated text features are then concatenated with the image features along the channel axis, producing a combined feature map of dimensions (1280x4x4).

The final classification network applies a single convolutional layer with a Sigmoid activation function to this combined feature map, reducing it to a scalar output, representing the probability that the input pair is real or generated.

3.3 Generator 2

The Generator 2 architecture leverages a combination of convolutional layers, residual blocks, and attention mechanisms to generate high-quality images from both image and text embeddings. The use of these advanced techniques ensures that the model is capable of generating coherent, contextually relevant images by

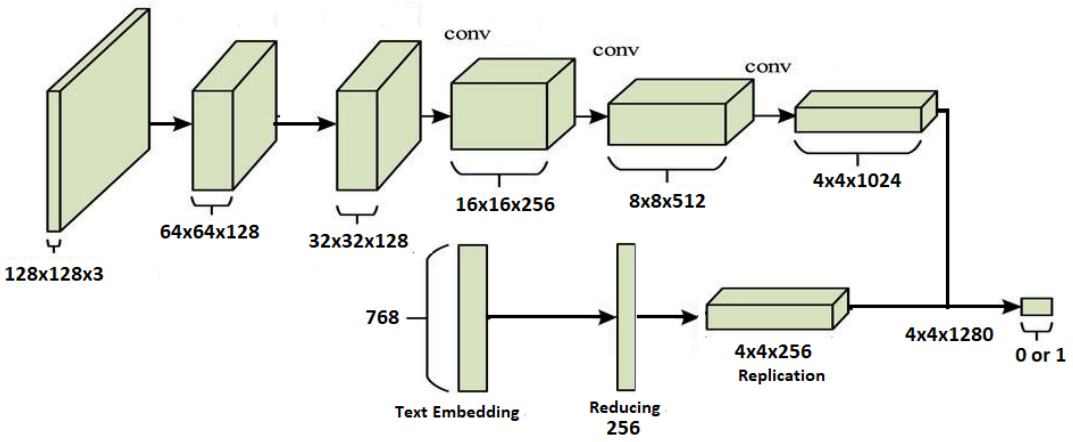


Figure 2: Discriminator 1: Processes an input image and text separately through image and text encoders. The extracted image features (1024×4×4) are concatenated with replicated text embeddings (256×4×4) before a final convolutional layer classifies the pair as real or generated.

efficiently combining visual and textual information.

Text embeddings of dimension 768 are passed through a fully connected layer, which reduces the dimensionality to 256. This process makes use of a linear transformation followed by Layer Normalization and a Leaky ReLU activation, allowing efficient handling of text features while avoiding feature collapse. The output is reshaped and expanded to (256x128x128) to match the spatial dimensions of the image input.

Input images of shape (3x128x128) are first processed by a 3x3 convolutional layer with feature size as 128. This is followed by Group Normalization and Leaky ReLU activation to ensure stable gradient flow and efficient feature encoding. A second 3x3 convolutional layer then increases the feature size to 256, producing an output of (256x128x128).

The Detail Branch focuses on enhancing high-frequency information critical for realistic image generation. An initial convolutional layer produces tensor of (512x128x128), followed by a Residual Block to preserve key details. This block consists of two 3x3 convolutional layers with Group Normalization, which helps retain relevant features from earlier layers. The output is further refined by a convolutional layer, reducing the feature size back to 256 and outputting (256x128x 128).

To condition image generation on textual information, we concatenate the projected text features and the enhanced image features, creating a tensor of (512x128x128). A 1x1 convolutional layer applies a spatial attention mechanism, producing attention weights in (256x 128x128) via a Sigmoid activation. These attention weights are used to scale the enhanced image features outputting (256x128x128).

The attended features are refined through a 3x3 convolutional layer to reduce feature dimensions to (128x128x128). This is followed by a second convolutional layer that outputs (channelsx128x128) using a Tanh activation, creating the final generated image. This refinement helps mitigate over-smoothing and enhances detail representation.

A learnable detail weight is calculated by applying an Adaptive Average Pooling layer to the attended features, followed by a 1x1 convolution, producing a scalar weight in (1x1x1). This weight is used to blend the original image input and the refined image output, producing the final output image. This residual connection allows fine-tuned blending, helping to retain relevant details from the input image.

3.4 Generator 3

The Generator 3 is based on the SRGAN architecture and is designed to upscale the output of Generator 2 from a resolution of 128x128 to 256x256. It begins with a 9x9 convolutional layer, which extracts shallow features from the input image, followed by a PReLU activation function for non-linear feature transformation.

The central processing involves 5 Residual Blocks, where each block includes two 3x3 convolutional layers, Batch Normalization (BatchNorm) for stabilizing feature distributions, and a skip connection to facilitate residual learning and retain important input information. This setup allows the network to focus on refining fine details and textures.

After the residual processing, the features pass through another 3x3 convolutional layer with BatchNorm. A skip connection is added here to merge the residual output with the shallow features, preserving both low-level and high-level details.

The Upsample Block then performs upsampling by a factor of 2, using Pixel Shuffle to double the

spatial resolution twice (from 128x128 to 256x256) while maintaining computational efficiency. Finally, a 9x9 convolutional layer refines the RGB output, and a Tanh activation function normalizes the pixel values. The final scaling factor (2.38 * output + 0.26) adjusts the pixel intensity to ensure perceptually realistic high-resolution outputs. This design effectively enhances the resolution while seamlessly integrating with the output from Generator 2.

3.5 Discriminator 2

Discriminator 2 in this setup is designed to classify images as real or generated, similar to Discriminator 1, but with a different architecture and focus. It operates on a 256x256 image and uses a series of convolutional layers to extract features. As the image passes through each convolutional block, its spatial dimensions are reduced while the depth of feature maps increases. By the time it reaches the final layer, the network has learned a rich representation of the image, which is then flattened and passed through a fully connected layer to output a scalar probability that indicates whether the image is real or fake. This design allows Discriminator 2 to efficiently capture both low- and high-level features from the input image.

Compared to Discriminator 1, which processes both image and text inputs, Discriminator 2 focuses solely on images. While Discriminator 1 aligns text features with the image and combines them for classification, Discriminator 2 operates in a more traditional GAN setup utlizing SRGAN specifically, relying entirely on the image for real/fake classification.

3.6 Dataset and Preprocessing

There are limited datasets available specifically for text-to-face synthesis, making it a challenging area for model training. Common face datasets, such as Labelled Faces in the Wild (LFW) [21], MegaFace [22], and VGGFace2, primarily focus on face recognition tasks and lack detailed attribute annotations or text descriptions. LFW contains labeled images for identity verification but has limited diversity in attributes. MegaFace offers a large-scale dataset for face recognition but does not provide annotations for finer facial traits that are essential for text-to-face generation.

We chose the CelebA [23] dataset for its comprehensive annotations and balance of dataset size with diverse facial attributes. CelebA includes 202,599 facial images, each annotated with 40 binary attributes, providing rich metadata essential for generating realistic, attribute-driven images from text.
We use the CelebA dataset with 20,000 annotated face images. Text captions are generated from facial attributes (e.g., "young female with black hair and glasses") and encoded using Sentence-BERT. We adopt the caption generation algorithm proposed by Deorukhkar et al. [2], which effectively maps structured attribute data into natural language sentences. This approach organizes attributes into six categories, allowing for the generation of diverse captions from a given list of attributes. This method enables the creation of N unique captions per image, providing varied textual descriptions for training.

For effective text-to-image synthesis, it's necessary to convert these textual descriptions into embeddings, a dense, continuous representation of semantic information. To achieve this, we used Sentence-BERT (SBERT) [10], a model designed for sentence-level embeddings, which captures nuanced meanings from complex attribute-based captions. SBERT transforms each caption into a high-quality embedding that retains the unique combination of attributes in the description, thereby enhancing the model's ability to generate images accurately aligned with the text.

3.7 Training Process

The main objective function used in GANs is:

$$\min_G \max_D V(D,G) = \min_G \max_D \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1-D(G(z)))] \quad (1)$$

where G = Generator, D = Discriminator, Pd (x)= distribution of real data, P(z) = distribution of generator, x = sample from Pd (x), z = sample from P(z), D(x) = Discriminator network, G(z) = Generator network

The training process involves three generators and two discriminators. For training Generator 1 and Generator 2, we use Discriminator 1, which is responsible for distinguishing between real and generated image-text pairs. Once the training for Generator 1 is complete, we will shift focus to Generator 2 and eventually Generator 3, for which we use Discriminator 2. Discriminator 2 is specifically designed to evaluate the outputs of Generator 3, ensuring that the progression of image quality and text alignment continues to improve as the process moves from Generator 1 to Generator 3.

3.7.1 Generator 1

The process begins by feeding the Generator 1 with two inputs: a random noise vector (latent vector) and a text embedding. The random noise introduces variability into the generated images, while the text embedding ensures that the generated images are conditioned on the provided textual description. Generator 1 is trained using Discriminator 1, which evaluates whether the generated image-text pair is real or fake.

The loss function for Generator 1 is a combination of adversarial loss, feature matching loss, and L1 loss.

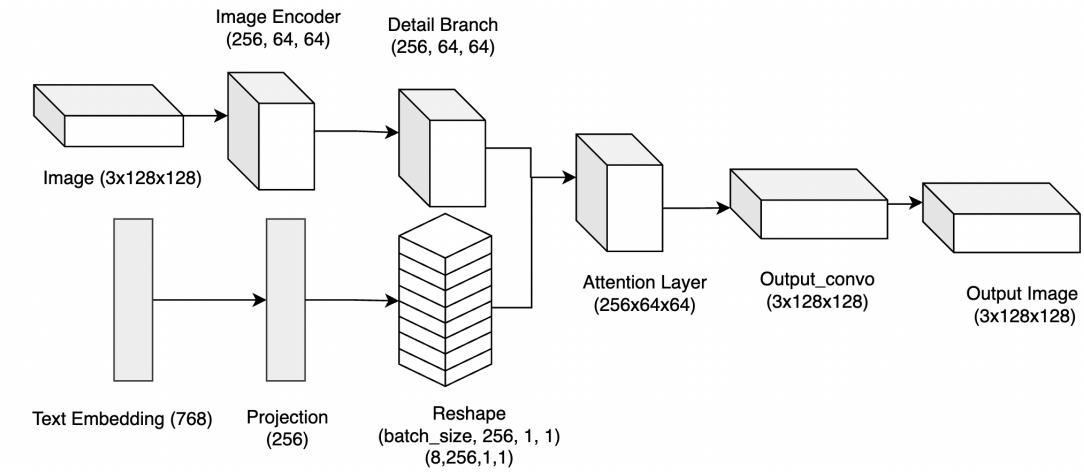


Figure 3: Generator 2: Combines convolutional layers, residual blocks, and attention mechanisms to generate high-quality images by integrating image and text embeddings. Text embeddings are projected and concatenated with image features, refined through spatial attention and convolutional layers, producing the final output with enhanced detail preservation.

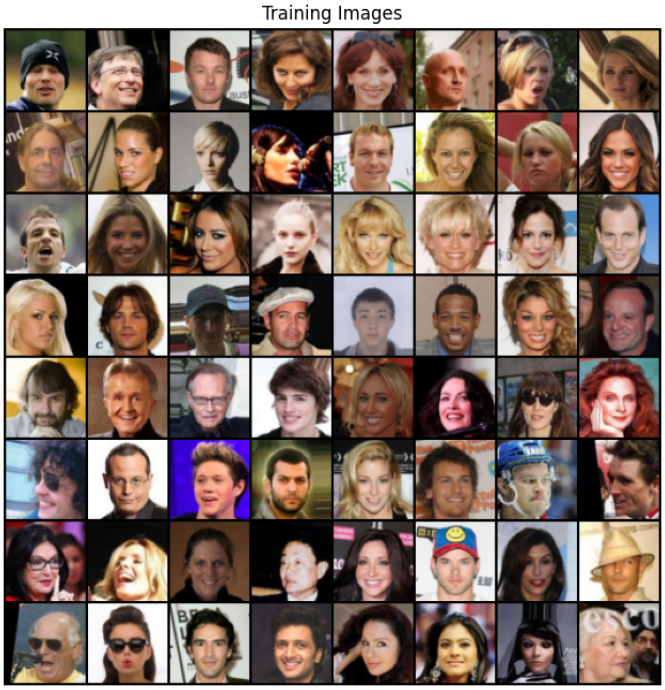


Figure 4: Sample frames from the CelebA dataset [23], which includes over 200,000 celebrity images annotated with facial attributes.

The primary objective of Generator 1 is to learn to generate images that are realistic and consistent with the textual descriptions provided. The adversarial loss encourages the generator to produce images that deceive the discriminator into classifying them as real. This loss is mathematically expressed as:

$$L_{\text{adv}} = -\mathbb{E}[\log D(G(z, c))] \tag{2}$$

where $G(z, c)$ represents the generated image conditioned on noise z and caption c, and D is the discriminator.

Feature matching loss ensures that the generated images align with real images at a feature level, encouraging the generator to produce images with meaningful structures. The feature matching loss is defined as:

$$L_{\text{fm}} = \mathbb{E}\left[\|\varphi(D(\hat{x})) - \varphi(D(x))\|_2^2\right] \tag{3}$$

where $\hat{x}$ is the generated image, x is the real image, and $\varphi(D(x))$ represents the features extracted from the discriminator's intermediate layers.

Additionally, L1 loss is used to minimize the pixel-wise difference between the generated and real images, helping the generator produce sharper and more realistic images. This is given by:

$$L_1 = \mathbb{E}\left[\|\hat{x} - x\|_1\right] \tag{4}$$

where $\hat{x}$ and x are the generated and real images, respectively.

The total loss function for Generator 1 combines these three components:

$$L_{\text{gen}} = L_{\text{adv}} + \lambda_{\text{fm}} L_{\text{fm}} + \lambda_{L_1} L_{L_1} \tag{5}$$

where $\lambda_{\text{fm}} = 100$ and $\lambda_{L_1} = 50$.
We have used $\lambda_{\text{fm}} = 100$ and $\lambda_{L1} = 50$

For optimization, Adam optimizer is used for both the generator and the discriminator. The learning rate for the optimizer is set to 0.0002, and the betas are set to $\beta_1 = 0.5$ and $\beta_2 = 0.999$ which are commonly used in generative adversarial network (GAN) training to stabilize the learning process.

3.7.2 Generator 2

The training process for Generator 2 builds upon the output of Generator 1, aiming to refine the generated images and ensure better alignment with textual descriptions. Generator 2 processes images of size 128x128 (output from Generator 1) and corresponding text embeddings. Its main goal is to improve image quality, add finer details, and enhance the semantic alignment between the image and text description.

Generator 2 consists of several stages designed to process the image and text embeddings efficiently. First, the text embeddings are projected into a reduced-dimensional space using a fully connected layer followed by Layer Normalization and a Leaky ReLU activation, allowing for efficient text feature processing. These features are then spatially expanded to align with the image dimensions, ensuring that the text is incorporated in the image generation process.

In parallel, the image input is passed through a series of convolutional layers with Group Normalization and Leaky ReLU activations to extract meaningful features while maintaining computational efficiency. The image features are then passed through a detail branch that enhances fine-grained details, incorporating a residual block to preserve and refine these details. To further align the image with the provided text, a lightweight attention mechanism is introduced. This mechanism combines the enhanced image features with the text embeddings, generating attention weights that are used to modulate the image features, ensuring that the output is better aligned with the text description. The final image is produced through additional convolutional layers in the output stage, using a tanh activation function to generate the final output image.

The loss function for Generator 2 consists of three main components.

Adversarial Loss The adversarial loss encourages Generator 2 to produce images that can deceive the discriminator:

$$L_{\text{adv}} = -\mathbb{E}\left[\log D\left(G_2(\hat{x}, c)\right)\right] \tag{6}$$

where $\hat{x}$ is the input image from Generator 1.

Content Loss The content loss minimizes the pixel-wise difference between the output of Generator 2 and its input:

$$L_{\text{content}} = \mathbb{E}\left[\|\hat{x}_{G_2} - \hat{x}\|_1\right] \tag{7}$$

where $\hat{x}_{G_2}$ is the output of Generator 2.

Detail Loss The detail loss focuses on preserving local variations and fine-grained details:

$$L_{\text{detail}} = \mathbb{E}\left[\|(\hat{x}_{G_2} - \text{pool}(\hat{x}_{G_2})) - (\hat{x} - \text{pool}(\hat{x}))\|_1\right] \tag{8}$$

where pool represents an average pooling operation.

Total Loss The total loss function is a weighted sum of these three losses, with the adversarial loss given a weight of 0.2, content loss given 0.6, and detail loss given 0.2. The total loss is expressed as:

$$L_{G_2} = \alpha L_{\text{adv}} + \beta L_{\text{content}} + \gamma L_{\text{detail}} \tag{9}$$

where $\alpha = 0.2$, $\beta = 0.6$, and $\gamma = 0.2$.

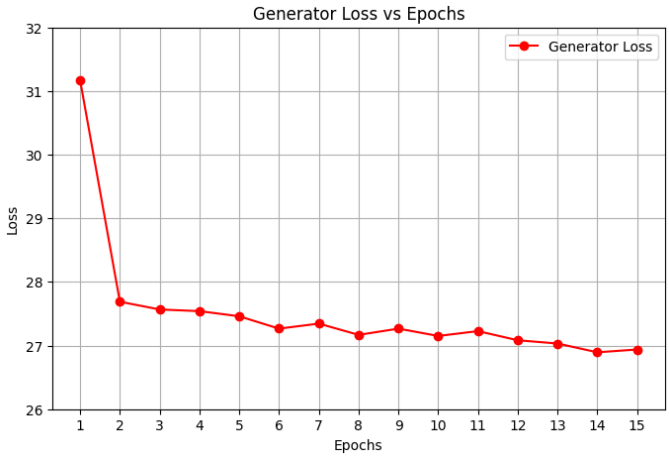


Figure 5: Generator 1: Loss vs Epoch curve over 20 training epochs.

3.7.3 Generator 3

The Generator 3 network is designed for single-image super-resolution[6]. This generator is tasked with enhancing the resolution of images from 128x128 pixels (output of Generator 2) to 256x256 pixels while maintaining visual fidelity and alignment with semantic content.The architecture includes residual blocks with skip connections to preserve spatial features and stabilize training, and upsampling blocks that use pixel shuffling to progressively increase resolution. The final output layer generates a high-resolution image normalized via Tanh activation.

The training process for Generator 3 employs multiple loss functions. Adversarial loss encourages the generator to produce realistic images that can deceive Discriminator 2, defined as

$$L_{\text{adv}} = -\mathbb{E}\left[\log D_2\left(G_3(z)\right)\right] \tag{10}$$

Additionally, perceptual loss leverages features from a pre-trained VGG network[6]. to ensure the generated images align semantically with real ones, further enhancing visual fidelity.

4 Result

The training process demonstrated effective convergence across all the stages. After 15 epochs, Generator 1's loss decreased from 31.236 to 26.984, and Discriminator 1's loss reduced from 3.783 to 1.545, indicating improved adversarial balance. Generator 2, trained for 20 epochs, exhibited a reduction in loss from 0.1229 to 0.1104, while Discriminator 2's loss similarly decreased from 0.6810 to 0.6572, suggesting improved stability and semantic alignment during the second generative stage.

The graphs in Figure 5 and Figure 6 illustrate the training progression of Generator 1 and Discriminator 1. The consistent decline in both loss curves reflects improved generation quality and adversarial learning stability.

Figure 7 shows the loss trends for Generator 2 and Discriminator 2. The discriminator loss (solid line) decreases steadily, while the generator loss (dashed line) remains low and stable, indicating high-quality image synthesis and effective adversarial learning.

The performance of the model was evaluated using standard generative metrics, namely Inception Score (IS) and Fréchet Inception Distance (FID). IS provides a measure of image quality and diversity, while FID assesses the similarity between real and generated image distributions. A higher Inception Score reflects both the quality and diversity of the generated images.A lower FID score indicates that the generated images are closer in distribution to the real images, reflecting higher realism. Given the limited class diversity in face datasets like CelebA, FID is considered a more reliable indicator of visual fidelity in this context.

For our model both FID and IS scores are computed using 20,000 images, with real and generated images sampled in batches of 8. These images are collected from a dataloader, which ensures that the evaluation uses a diverse and representative set of images for both the real and generated datasets. The use of 20,000 samples provides a robust estimate of the performance of the generators.
Our results are summarized in Table II.

Metric	**T2FGAN**
Inception Score (IS)	1.42 ± 0.03
FID Score	96.97

Table 2 Quantitative evaluation metrics for the proposed T2FGAN model.

The model achieved an FID score of 96.97 and an IS of 1.42 ± 0.03. While the FID score indicates room

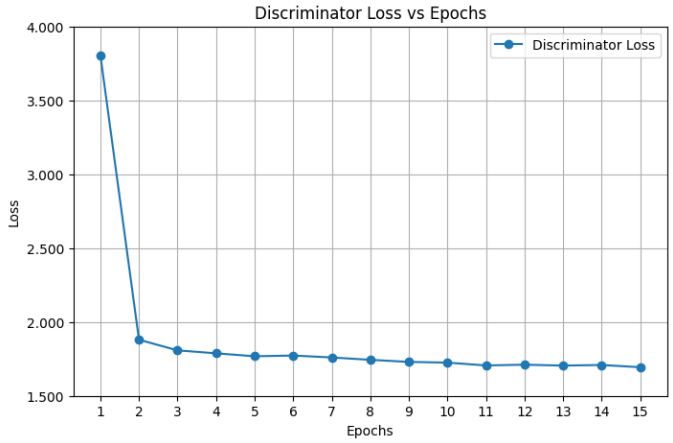


Figure 6: Discriminator 1: Loss curve over 15 training epochs, showing its ability to distinguish real and generated images.



Figure 7: Generator 2 and Discriminator 2: Training loss curves over 20 epochs.

for improvement in aligning generated images with the real data distribution, the IS reflects moderate image diversity and semantic coherence—appropriate given the limited class variation in the CelebA dataset.

It is important to contextualize these results in light of the training configuration and dataset size. Due to CUDA memory limitations, Generator 1 was trained on 20,000 images, while Generator 2 was trained on only 4,000 images. Under these constraints, the model achieved an FID score of 96.97. The reduced dataset size and limited training duration likely contributed to the current performance gap. With extended training and access to a larger dataset, the performance of T2FGAN is expected to improve significantly, with the potential to surpass existing benchmarks.

In order to further evaluate the performance of our model, we present a qualitative comparison of the generated images against the ground truth in Table 3

5 Future Scope

In future work, several improvements could be made to further enhance the performance and versatility of our model. First, implementing a more advanced dataset balancing strategy could help address the issue of varying text description lengths, from very short to highly detailed. By accounting for these differences more effectively, the model could generalize better to a wider range of input styles, ultimately improving its flexibility and robustness.

Currently, our model is trained for 20 epochs, yielding promising results. However, increasing the number of training epochs would likely enhance the model's ability to learn more intricate relationships between the textual and visual domains, leading to more accurate and detailed image generation. Longer training periods may allow the model to fine-tune its understanding of subtle patterns in the data, further improving the output quality.

Another potential improvement involves extending the resolution of the generated images to 512x512, or even higher. This would allow the model to produce more detailed and realistic images, capturing finer nuances that are crucial for high-quality image synthesis. However, this would also necessitate modifications to the model architecture and require additional computational resources to handle the increased complexity.

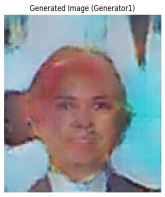
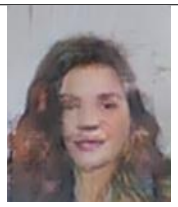
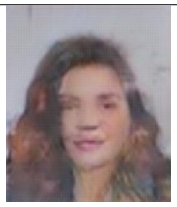
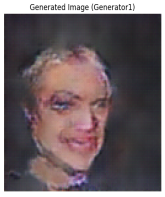
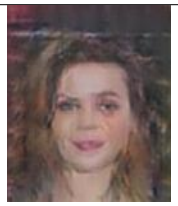
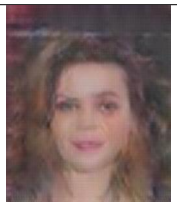
Sentence	Ground Truth	Generated Image1	Generated Image2	Generated Image3
The male has an oval face. He has sideburns. His hair is black. The gentleman is smiling, is attractive and young.				
Her hair is brown, wavy and straight. She has arched eyebrows, big lips, a slightly open mouth and a pointy nose. The woman is attractive, young and has heavy makeup. She is wearing lipstick.				
The woman has high cheekbones. She has brown and wavy hair. She has arched eyebrows, bushy eyebrows and a pointy nose. She is smiling, looks attractive, young and has heavy makeup. She is wearing earrings and lipstick.				

Table 3 Comparison of Ground Truth and Generated Images from Generator 1, Generator 2, and Generator 3. This table presents qualitative results of text-to-image generation, where each row corresponds to a specific descriptive text input. The first column contains textual descriptions, while the second column shows real images (Ground Truth) that match the given description. The subsequent columns display images generated by three different models—Generator 1, Generator 2, and Generator 3. The comparison highlights the variations in facial structure, attributes, and realism produced by each generator, showcasing improvements in fine-grained details, texture quality, and adherence to the input description across different approaches.

Additionally, integrating multimodal learning—training models on text, images, and other data like audio or video—could enhance contextual understanding and improve image generation accuracy. This approach would enable richer feature extraction, making models more adaptable to complex or abstract descriptions.

Future advancements in text-to-image generation could benefit from transformer-based models, diffusion models like Stable Diffusion, and CLIP-guided models like DALL·E [24]. These approaches offer improved text-image alignment and semantic accuracy, making them promising alternatives to GANs. Integrating such models could enhance fine-grained control over generated images, enabling more adaptable and realistic synthesis for diverse real-world applications.

Together, these future improvements — from better dataset balancing and extended training to higher resolution images, multimodal learning, and transformer-based approaches — offer promising avenues for enhancing text-to-image generation models. These changes would not only expand the model's applicability but also improve the realism and accuracy of generated images across various domains.

6 Conclusion

In this paper, we introduced T2FGAN, a novel model combining DCGAN and SRGAN with an attention mechanism to generate human faces from textual descriptions. The model employs a multi-stage architecture with three generators and two discriminators, allowing for progressive refinement in face generation and aiming to enhance the overall quality and diversity of the generated images.

While the comparison with DCGAN in terms of Inception Score (IS) and Fréchet Inception Distance (FID) did not show significant improvements, we observed a clear progression in the quality of the images through the different stages of the model. This suggests that the multi-stage refinement process helps improve the realism and alignment of the generated faces over time, even if the overall scores are not as high as those

achieved by DCGAN. Additionally, it is important to note that our model was trained for only 20 epochs across all three generators. In contrast, many researchers have trained similar models for significantly more epochs, which may explain the observed performance difference.

In conclusion, T2FGAN presents a promising approach for text-to-face generation, demonstrating that multi-generator and multi-discriminator architectures can progressively improve the generation process. While this work shows promising results, the exploration of more complex and diverse textual inputs remains an area for future research. This model opens up exciting opportunities for applications in personalized content generation and human-computer interaction, while also raising important ethical considerations about the responsible use of such technologies in generating realistic human faces.

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8 Statements and Declarations

The authors have no competing interests to declare that are relevant to the content of this article.

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